

Introduction to Applied Cryptography – LDACS1310

Exercises Session 1

Reminder of probability concepts

Expected value Calculates the mean.

$$\mathbb{E}[X] = \sum_i x_i \cdot \Pr[X = x_i]$$

Variance σ^2 and Standard deviation σ Measures the dispersion around the expected value.

$$\begin{aligned} \text{Var}[X] &= \sigma^2 = \mathbb{E}[X - \mathbb{E}[X]]^2 = \mathbb{E}[X^2] - \mathbb{E}[X]^2 \\ \sigma &= \sqrt{\text{Var}[X]} \end{aligned}$$

Bayes' theorem Calculates the conditional probability that an event A occurring given that B is true (posterior probability).

$$\Pr[A = a|B = b] = \frac{\Pr[B = b|A = a] \cdot \Pr[A = a]}{\Pr[B = b]}$$

where $\Pr[B = b|A = a]$ is the conditional probability of event B occurring given that A is true, and $\Pr[A = a]$ and $\Pr[B = b]$ are the probability of observing A and B respectively.

Law of total probability Calculates the probability of an event by taking into account all the possible cases in which it may occur.

$$\Pr[A = a] = \sum_B \Pr[A = a|B = b'] \cdot \Pr[B = b']$$

Questions

Question 1: Two-Key Pad

Let $\Pi_{otp1} = \langle Gen, Enc, Dec \rangle$ be an encryption scheme, where Gen and Enc are defined as follow:

$\text{Gen}(1^\lambda)$
 $k_0, k_1 \leftarrow \{0, 1\}^n$
 $sk = (k_0, k_1)$

$\text{Enc}_{sk}(m), m \in \{0, 1\}^n$
 $m = m_1 || m_2 || \dots || m_n$
 $c = m \oplus k_{m_n}$

- (a) Prove that this scheme satisfies perfect secrecy as seen in class.
- (b) This scheme is not correct. Some ciphertexts can not be decrypted under some keys. Show an example in which decryption is impossible. What is the probability of decryption failing ?

Question 2: Two-Key Pad - Fixed

Let $\Pi_{otp2} = \langle Gen, Enc, Dec \rangle$ be an encryption scheme, where Gen and Enc are defined as follow:

$Gen(1^\lambda)$	$Enc_{sk}(m), m \in \{0, 1\}^n$
$k_0, k_1 \leftarrow \{0, 1\}^{n-1}, b \leftarrow \{0, 1\}$	$m = m_1 m_2 \dots m_n = m^* m_n$
$sk = (k_0, k_1, b)$	$c^* = m^* \oplus k_{m_n}$
	$c_n = b \oplus m_n$
	$c = c^* c_n$

- Give the Dec algorithm to make Π_{otp2} a correct scheme.
- Show that this scheme is secure
- This scheme draws $2n$ bits of random value to encrypt n bits of message. Is this scheme 2-times secure ?

Question 3: Some Negligible Functions Properties

Remember that a function $f : \mathbb{N} \rightarrow \mathbb{R}^+$ is negligible if for every positive polynomials $p(\cdot)$ there is an N such that for all integers $n > N$, $f(n) < \frac{1}{p(n)}$.

- Let f_1 and f_2 be negligible functions, show that $f_3 = f_1 + f_2$ is negligible.
- Let f_1 be a negligible function, for any positive polynomial p show that $f_2 = p(n) \cdot f_1(n)$ is negligible.
- There are more similar properties. For any constant $c > 0$, if f is a negligible function, then $c \cdot f$ and f^c are negligible functions as well. Why are those composition properties useful in cryptography ?

Question 4: Overwhelming Function

The complement of negligible functions is called overwhelming functions. Formally, a function f is overwhelming if $1 - f$ is negligible. In the following examples, would you want the probability to be negligible or overwhelming ?

- the probability that no successful attack takes place
- $\Pr[Dec_{sk}(Enc_{sk}(m)) = m]$
- the advantage of an adversary over an encryption scheme
- the probability of producing a signature forgery

Question 5: Random Permutation Cipher

Let Φ be the set of all permutations over $\{0, 1\}^n$, i.e. $\Phi = \{\phi : \{0, 1\}^n \rightarrow \{0, 1\}^n \mid \phi \text{ is a bijection}\}$. Remember that a bijection is a one-to-one correspondence between two sets, or said otherwise it is an invertible function. Let $\Pi_\Phi = \langle Gen, Enc, Dec \rangle$ be an encryption scheme defined as follow:

$Gen(1^\lambda)$	$Enc_{sk}(m), m \in \{0, 1\}^n$	$Dec_{sk}(c)$
$\phi \leftarrow \Phi$	$c = \phi(m)$	$m = \phi^{-1}(c)$
$sk = (\phi, \phi^{-1})$		

- Is this scheme correct ?
- Is this scheme secure ?
- Is this scheme 2-times secure ?
- What is the drawback of this scheme which makes it unusable ?